



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2007
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #1

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 1.5 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Question is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 76

- Attempt questions 1 – 3

Examiner: *P.Bigelow*

QUESTION 1 (23 marks)

Marks

(a) Find

(i) $\int x \sec^2(1+x^2) dx$ [2]

(ii) $\int x^2 e^{2x} dx$ [2]

(b) Evaluate

(i) $\int_0^1 \frac{dx}{(1+x)\sqrt{1+x}}$ [2]

(ii) $\int_0^1 \frac{3x+7}{(x+1)(x+2)(x+3)} dx$ [3]

(c)

(i) Show that $\frac{x^m + x^{m+2}}{1+x^2} = x^m$ [1]

(ii) An integral is defined by

$$I_m = \int_0^1 \frac{x^m}{1+x^2} dx$$

where m is a non-negative integer.

$$(\alpha) \quad \text{Evaluate } I_0. \quad [1]$$

$$(\beta) \quad \text{Show that } I_m + I_{m+2} = \frac{1}{m+1}. \quad [2]$$

$$(\gamma) \quad \text{Hence or otherwise evaluate } I_2. \quad [1]$$

(d) Five couples sit at a round table. How many seating arrangements are possible if

(i) there are no restrictions? [2]

(ii) each person sits next to their partner? [2]

(e) Let $z = \frac{1-i}{2+i}$. Find

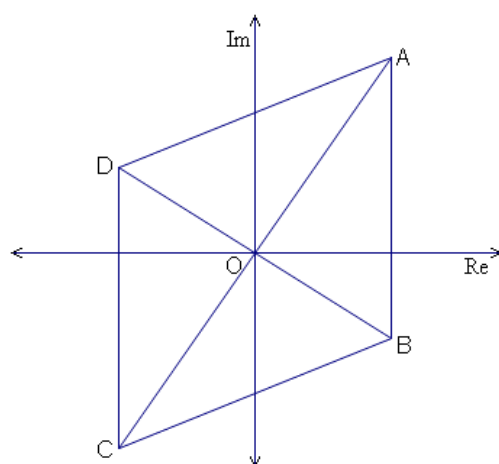
(i) $z + \frac{1}{z}$ [1]

(ii) $\bar{z}$ [1]

(f) ABCD is a rhombus, where diagonals meet at O. If A represents the complex number $3 + 4i$ and $OD = 2$ units find the complex number represented by

(i) C [1]

(ii) D [2]

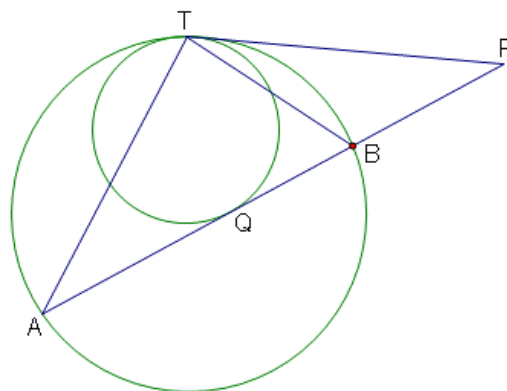


End of Question 1

QUESTION 2 (26 marks)

Marks

- (a) Sketch the following on separate Argand Diagrams.
- (i) $|z + 1| = 3$ [1]
- (ii) $\arg(z - 1) = \frac{\pi}{4}$ [1]
- (iii) $|z - 1| = \operatorname{Re} z$ [2]
- (iv) $\operatorname{Re}(z(\bar{z} + 2)) = 3$ [2]
- (b) Find the greatest and least values of $\arg z$ for which $|z - 4i| = 2$. [2]
- (c)
- (i) Express $1 - \sqrt{3}i$ in mod-arg form. [1]
- (ii) Hence find $(1 - \sqrt{3}i)^6$ in the form $a + bi$. [1]
- (d) PT is a common tangent to the circles which touch at T. PA is a tangent to the smaller circle at Q.
- (i) Prove that $\triangle BTP$ is similar to $\triangle TAP$. [2]
- (ii) Hence show that $TP^2 = PA \cdot PB$ [1]
- (iii) If $PT = t$, $QA = a$ and $QB = b$ prove that $t = \frac{ab}{a - b}$. [2]



- (e) ω is a complex cube root of unity. Evaluate $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$. [2]
- (f) If α, β and γ are roots of $2x^3 - 7x^2 + 5x - 3 = 0$
- (i) show that the equation with roots α^2, β^2 and γ^2 is $4x^3 - 29x^2 - 17x - 9 = 0$. [2]
- (ii) Hence or otherwise evaluate $\alpha^3 + \beta^3 + \gamma^3$. [2]
- (g) The equation $x^3 - 3px + q = 0$ (where $p > 0$, $q \neq 0$ are both real) has three distinct non-zero real roots.
- (i) Show that the graph $y = x^3 - 3px + q$ has a relative maximum of $q + 2p\sqrt{p}$ and a relative minimum of $q - 2p\sqrt{p}$. [3]
- (ii) Hence show that $q^2 < 4p^3$. [2]

End of Question 2

QUESTION 3 (27 marks)

Marks

- (a) The letters of KAYAK are arranged at random in a line.
- (i) How many different letter sequences are possible? [2]
- (ii) What is the probability that the sequence is the same from right to left as left to right? [2]
- (b)
- (i) Show that $z^5 + 1 = (z + 1)(1 - z + z^2 - z^3 + z^4)$. [1]
- (ii) Find in the form $\text{cis}\theta$, the five complex solutions to $z^5 + 1 = 0$. [2]
- (iii) If z is a solution to $z^5 + 1 = 0$ and $z \neq -1$, prove that $1 + z^2 + z^4 = z + z^3$. [1]
- (iv) Hence show that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} - \frac{1}{2} = 0$. [3]
- (c)
- (i) find the values of the constants A and B such that $4x^4 + 1 \equiv (2x^2 + Ax + 1)(2x^2 + Bx + 1)$. [2]
- (ii) Hence express the integer $2^{14} + 1$ as a product of prime factors. [2]
- (d) Solve $2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$ if it has a root of multiplicity 3. [2]
- (e)
- (i) Prove $\int_0^c f(x)dx = \int_0^c f(c-x)dx$. [2]
- (ii) Hence or otherwise find $\int_0^1 x(1-x)^{2007} dx$. [2]

- (f) The diagram shows an isosceles triangle PAB. PM is the perpendicular bisector of $\angle APB$, which is β . PM bisects AB. A and B represent the complex numbers z_1 and z_2 respectively.

- (i) Find the complex number represented by

(α) $\overrightarrow{AM}$

[1]

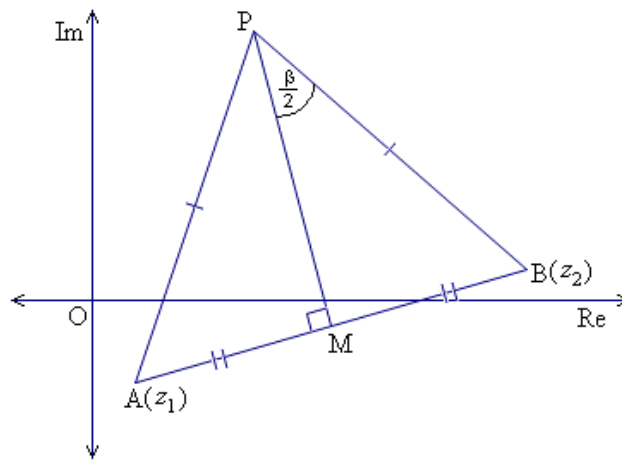
(β) $\overrightarrow{MP}$

[2]

- (ii) Hence show that P represents the complex number

$$\frac{1}{2} \left(1 - i \cot \frac{\beta}{2} \right) z_1 + \frac{1}{2} \left(1 + i \cot \frac{\beta}{2} \right) z_2$$

[3]



End of Exam

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Solutions to Question (1)

(a) (i) $\int x \sec^2(1+x^2) dx$
 $= \frac{1}{2} \tan(1+x^2) + c$ (2)

(ii) $\int x^2 \frac{d}{dx} \left(\frac{1}{2} e^{2x} \right) dx$
 $= \frac{x^2 e^{2x}}{2} - \int x e^{2x} dx$

Now $\int x e^{2x} dx = \int x \frac{d}{dx} \left(\frac{1}{2} e^{2x} \right) dx$

$= \frac{x e^{2x}}{2} - \frac{1}{4} \int 2 e^{2x} dx$

$= \frac{x e^{2x}}{2} - \frac{e^{2x}}{4}$

$\therefore \int x^2 e^{2x} dx$ (2)

$= \frac{x^2 e^{2x}}{2} - \frac{x e^{2x}}{2} + \frac{1}{4} e^{2x} + c$

$= \frac{e^{2x}}{4} (2x^2 - 2x + 1) + c$

$\frac{1}{2} x e^{2x} (x-1) + \frac{1}{4} e^{2x} + c$

(b) $\int_0^1 (1+x)^{-1/2} dx$

(i) $= \left[-2(1+x)^{1/2} \right]_0^1$

$= -2 \cdot \frac{1}{\sqrt{2}} + 2$ (2)

$= 2 - \sqrt{2} \quad (0.586)$

(ii) $\frac{3x+7}{(x+1)(x+2)(x+3)}$ (3)

$= \frac{a}{x+1} + \frac{b}{x+2} + \frac{c}{x+3}$

$\therefore 3x+7 = a(x+2)(x+3) + b(x+1)(x+3) + c(x+1)(x+2)$

put $x = -1, a = 2$

$x = -2, b = -1$

$x = -3, c = -1$

$\therefore \int \frac{(3x+7) dx}{(x+1)(x+2)(x+3)}$

$= \left[2 \ln(x+1) - \ln(x+2) - \ln(x+3) \right]_0^1$
 $= \ln 2 \quad (0.693)$

(c) (i) $\frac{x^m (1+x^2)}{1+x^2} = x^m$ (1)

(ii) $I_0 = \int_0^1 \frac{1}{1+x^2} dx$

$= \left[\tan^{-1} x \right]_0^1 = \frac{\pi}{4}$ (1)

(B) $I_m + I_{m+2} = \int_0^1 \frac{x^m}{1+x^2} dx$

$+ \int_0^1 \frac{x^{m+2}}{1+x^2} dx$

$= \int_0^1 \left(\frac{x^m + x^{m+2}}{1+x^2} \right) dx$ (2)

$= \int_0^1 x^m dx = \left[\frac{x^{m+1}}{m+1} \right]_0^1$
 $= \frac{1}{m+1}$

(8) $I_0 + I_2 = 1$

$\frac{\pi}{4} + I_2 = 1$ (1)

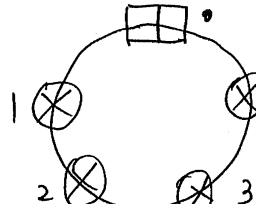
$\therefore I_2 = 1 - \frac{\pi}{4}$

$= \frac{4-\pi}{4} \quad (0.215)$

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(i) 9! (2)

(d) "Fix a couple for reference on a circle, the other couples can be arranged 4! ways"



each partner of each couple can swap without altering the arrangement

$\therefore \text{Total arrangement} = 4! \times 2^5 \quad (768)$

(e) $\frac{x-2i}{(2+i)(1-i)}$

$= \frac{3+2i}{3-i} \times \frac{(3+i)}{(3+i)}$

$= \frac{9+9i-2}{10}$

$= \frac{7+9i}{10}$ (1)

(ii) $\overline{z} = \frac{1+i}{2-i} \times \frac{(2+i)}{(2+i)}$

$= \frac{1+3i}{5}$ (1)

(f) $\overrightarrow{OA} = z_1$

(i) $= 3+4i$

$\therefore \overrightarrow{OC} = -\overrightarrow{OA}$

$= -3-4i$ (1)

$\therefore C = -3-4i$

$|\overrightarrow{OB}| = 2$

(ii) $\overrightarrow{OD} = \frac{2}{5} \overrightarrow{OA} \cos \frac{\pi}{2}$

$= \frac{2}{5} [(3+4i)i]$

$= \frac{2}{5} (-4+3i)$

$= -\frac{8}{5} + \frac{6i}{5}$ (2)

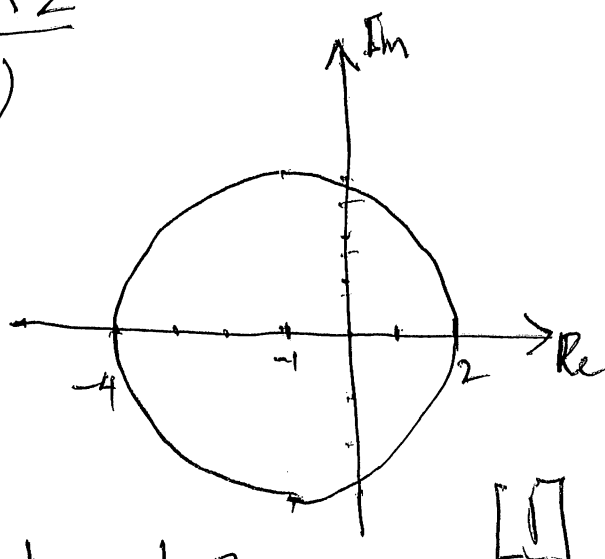
$\overrightarrow{OD} = 2 \cos \left(\frac{\pi}{2} + \tan^{-1} \frac{4}{3} \right)$

$[Total = 23]$

4 marks

Question 2

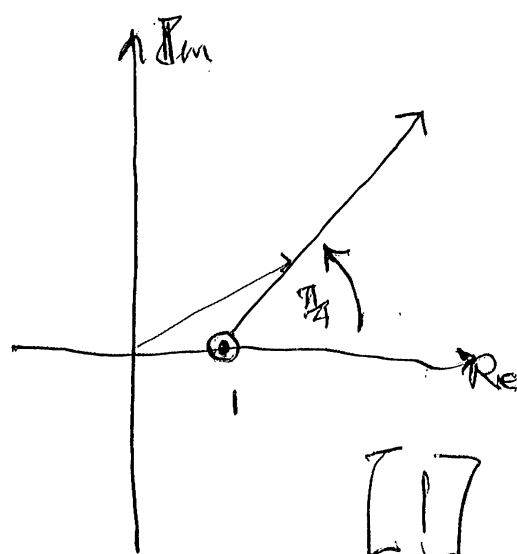
(a) (i)



$$|z+1|=3$$

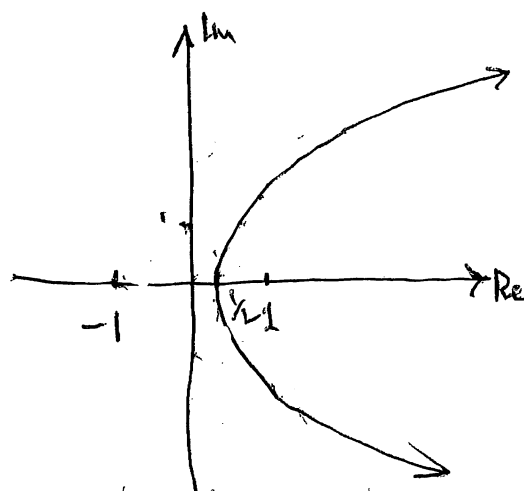
[1]

(ii)



[1]

(iii)



$$|z-1| = \operatorname{Re} z$$

$$|(x-1) + iy| = x$$

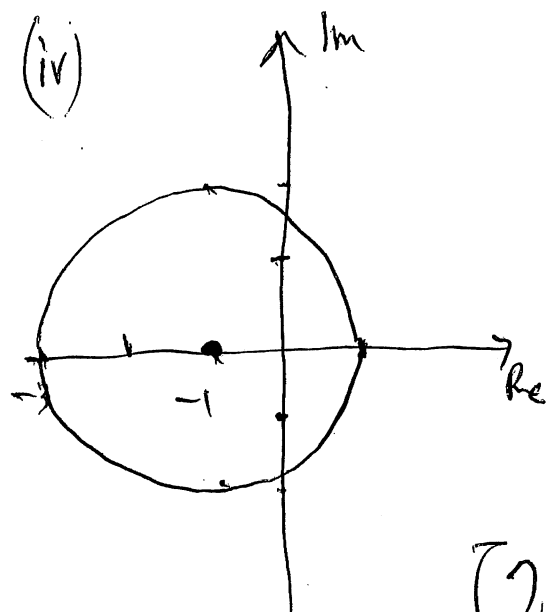
$$x^2 - 2x + 1 + y^2 = x^2$$

$$y^2 = 2x - 1$$

$$y = \pm \sqrt{2x-1}$$

[2]

(iv)



[2]

$$\operatorname{Re}(z(\bar{z}+2)) = 3$$

$$\operatorname{Re}(z\bar{z} + 2z) = 3$$

$$\operatorname{Re}(x^2 + y^2 + 2x + 2iy) = 3$$

$$x^2 + y^2 + 2x = 3$$

$$(x+i)^2 + y^2 = 4$$

$$(b) |z - 4i| = 2$$

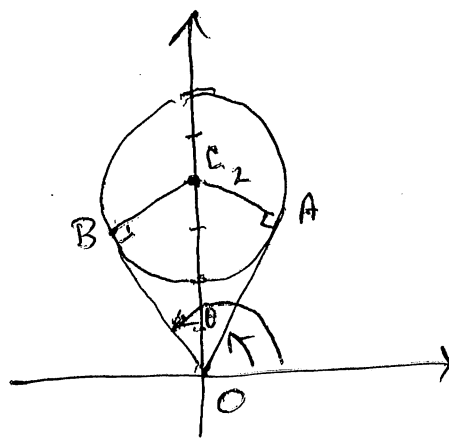
$$\text{In } \triangle OAC \quad \sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}$$

Hence

$$\frac{\pi}{3} \leq \arg z \leq \frac{2\pi}{3}$$

$$\therefore \text{Greatest value} = \frac{2\pi}{3}, \text{Least value} = \frac{\pi}{3} \quad [2]$$



(c)

$$(i) z = 1 - \sqrt{3}i$$

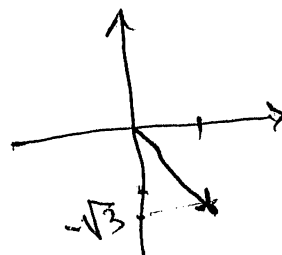
$$|z| = \sqrt{1+3}$$

$$= 2$$

$$\arg z = \tan^{-1} \frac{y}{x}$$

$$= \tan^{-1} \frac{-\sqrt{3}}{1}$$

$$= -\frac{\pi}{3}$$



$$\therefore z = 2 \operatorname{cis}\left(-\frac{\pi}{3}\right)$$

[1]

$$(ii) z^6 = \left(2 \operatorname{cis}\left(-\frac{\pi}{3}\right)\right)^6$$

$$= 64 \operatorname{cis}\left(-6 \times \frac{\pi}{3}\right)$$

$$= 64 \operatorname{cis}(-2\pi)$$

$$= 64 \operatorname{cis} 0$$

$$\therefore (1 - \sqrt{3}i)^6 = 64 + 0i$$

[1]

(d)(i) In the Δ 's BTP, TAP

$\angle BTP = \angle TAP$ (alternate segment theorem)

$\angle P$ is common

$\therefore \triangle BTP \parallel \triangle TAP$ (equiangular)

[2]

(d)(i) $\frac{PB}{PT} = \frac{PT}{PA}$ (corresponding sides of similar triangles)

[1]

$\therefore PA \cdot PB = PT^2$

(iii) $PT = PQ$ (tangents from an external point are equal.)

$\therefore PB = t - b, PA = t + a$

Thus $(t - b)(t + a) = t^2$ (from above)

$t^2 + ta - tb - ab = t^2$

$t(a - b) = ab$

[2]

$t = \frac{ab}{a - b}$ QED

(e) Since $1, w, w^2$ are roots of $z^3 - 1 = 0$,
 $1 + w + w^2 = 0$ (sum of roots)
 $\therefore 1 + w = -w^2$

Thus $\text{Exp} = (1 - 3w + w^2)(1 + w - 8w^2)$
 $= (1 - 3w - 1 - w)(-w^2 - 8w^2)$
 $= (-4w)(-9w^2)$
 $= 36w^3$
 $= 36 \quad (w^3 = 1)$

[2]

$$(f)(i) 2x^3 - 7x^2 + 5x - 3 = 0$$

Transformation: Let $y = x^2$
 $\therefore x = \sqrt{y}$

$$2y\sqrt{y} - 7y + 5\sqrt{y} - 3 = 0$$

$$\sqrt{y}(2y + 5) = 7y + 3$$

Square Both sides

$$y(4y^2 + 20y + 25) = 49y^2 + 42y + 9$$

$$\therefore 4y^3 - 29y^2 - 17y - 9 = 0$$

$\therefore$ Eq'n in x with roots $\alpha^2, \beta^2, \gamma^2$

$$4x^3 - 29x^2 - 17x - 9 = 0$$

[2]

(ii) Since α, β, γ are zeroes

$$2\alpha^3 - 7\alpha^2 + 5\alpha - 3 = 0$$

$$2\beta^3 - 7\beta^2 + 5\beta - 3 = 0$$

$$2\gamma^3 - 7\gamma^2 + 5\gamma - 3 = 0$$

By addition

$$2(\alpha^3 + \beta^3 + \gamma^3) - 7(\alpha^2 + \beta^2 + \gamma^2) + 5(\alpha + \beta + \gamma) - 9 = 0$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = \frac{1}{2} \left[7\left(\frac{29}{4}\right) - 5\left(\frac{7}{2}\right) + 9 \right]$$

$$= \frac{169}{8}$$

[2]

$$x^3 - 3px + q = 0$$

$p > 0, q \neq 0$, real
— distinct non-zero
real roots

$$(1) \quad y = x^3 - 3px + q$$

$$y' = 3x^2 - 3p$$

$$y' = 0 \text{ when } x^2 - p = 0$$

$$x = \pm\sqrt{p}$$

$$y'' = 6x$$

$$\text{At } x = \sqrt{p} \quad y'' > 0$$

$\therefore$ Relative minimum

$$\begin{aligned} y(\sqrt{p}) &= p\sqrt{p} - 3p\sqrt{p} + q \\ &= q - 2p\sqrt{p} \end{aligned}$$

$$\text{At } x = -\sqrt{p}$$

$$y'' < 0$$

$\therefore$ Relative maximum

$$\begin{aligned} y(-\sqrt{p}) &= -p\sqrt{p} + 3p\sqrt{p} + q \quad [3] \\ &= q + 2p\sqrt{p} \end{aligned}$$

(11) Since cubic has 3 non-zero real roots,
the turning points lie on opposite
sides of $y=0$.

$\therefore$ They are of different signs.

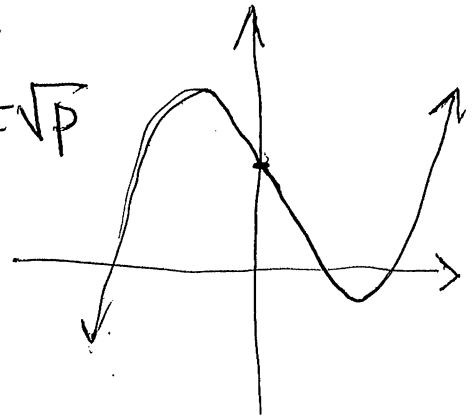
$\therefore$ Their product is < 0 .

$$\therefore (q + 2p\sqrt{p})(q - 2p\sqrt{p}) < 0$$

$$\therefore q^2 < 4p^3$$

[2]

Q.E.D.



$$3(a)(i) \frac{5!}{2!2!} = 30 \quad (2)$$

$$(ii) \frac{2}{30} = \frac{1}{15} \quad (2)$$

$$(b)(i) (z+1)(1-z+z^2-z^3+z^4) \\ = z - z^2 + z^3 - z^4 + z^5 + 1 - z + z^2 - z^3 + z^4 \\ = z^5 + 1 \quad (1)$$

$$(ii) (\cos \theta)^5 + 1 = 0 \\ \cos 5\theta = -1$$

$$\cos 5\theta = -1 \quad \sin 5\theta = 0$$

$$-\pi < \theta \leq \pi$$

$$-\pi < 5\theta \leq 5\pi$$

$$\therefore 5\theta = -3\pi, -\pi, \pi, 3\pi, 5\pi$$

$$\therefore \theta = -\frac{3\pi}{5}, -\frac{\pi}{5}, \frac{\pi}{5}, \frac{3\pi}{5}, \pi$$

$\therefore$ Solutions are

$$\cos(-\frac{3\pi}{5}), \cos(-\frac{\pi}{5}), \cos\frac{\pi}{5}, \cos\frac{3\pi}{5}, \cos\pi \quad (2)$$

$$(iii) z^5 + 1 = 0 \\ (z+1)(1-z+z^2-z^3+z^4) = 0 \\ \therefore 1-z+z^2-z^3+z^4 = 0 \quad (\text{as } z \neq -1) \\ \therefore 1+z^2+z^4 = z+z^3 \quad (1)$$

$$(iv) \text{ Consider } z = \cos \frac{\pi}{5} \\ 1+z^2+z^4 = z+z^3 \\ 1+\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = \cos \frac{\pi}{5} + \cos \frac{3\pi}{5}$$

Equating real parts

$$1 + \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} \\ \therefore 1 + \cos(\pi - \frac{3\pi}{5}) + \cos(\pi - \frac{\pi}{5}) = \cos \frac{\pi}{5} + \cos \frac{3\pi}{5}$$

$$\therefore 1 - \cos \frac{3\pi}{5} - \cos \frac{\pi}{5} = \cos \frac{\pi}{5} + \cos \frac{3\pi}{5}$$

$$\therefore 2 \cos \frac{\pi}{5} + 2 \cos \frac{3\pi}{5} = 1 = 0$$

$$\therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} - \frac{1}{2} = 0 \quad (3)$$

$$(c)(i) (2x^2+Ax+1)(2x^2+4x+1) \\ = 4x^4 + 2Bx^3 + 2x^2 + 2Ax^3 + ABx^2 \\ + Ax + 2x^2 + Bx + 1 \\ = 4x^4 + (2B+2A)x^3 + (AB+4)x^2 \\ + (A+B)x + 1$$

$$\therefore A+B=0 \Rightarrow B=-A$$

$$AB+4=0$$

$$\therefore -A^2+4=0$$

$$\therefore A^2=4$$

$$\therefore A = \pm 2 \quad B = \mp 2 \quad (2)$$

$$(ii) 2^{4+1} \\ = 4 \cdot 2^{12} + 1 \Rightarrow x = 2^3 = 8 \\ = (128+16+1)(128-16+1) \\ = 145 \times 113 \\ = 5 \times 29 \times 113 \quad (2)$$

$$(d) 2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$$

$$P'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$P''(x) = 24x^2 + 54x + 12$$

$$\text{Consider } 24x^2 + 54x + 12 = 0$$

$$\therefore 4x^2 + 9x + 2 = 0$$

$$\therefore (4x+1)(x+2) = 0$$

$$\therefore x = -\frac{1}{4} \text{ or } x = -2$$

$$P(-2) = 32 - 72 + 24 + 40 - 24 \\ = 0$$

$\therefore -2$ is the triple root.

$$\text{Sum of roots: } -6 + \beta = -\frac{9}{2}$$

$$\therefore \beta = 1\frac{1}{2}$$

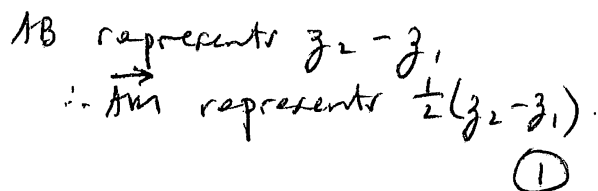
$\therefore$ Solutions are -2 and $1\frac{1}{2}$
 $\uparrow$
 triple root.

(2)

(iii) $\therefore P$ represents

$$= \left(\frac{1}{2} - i \cos t \frac{p}{2}\right) \beta_1 + \left(\frac{1}{2} + i \cos t \frac{p}{2}\right) \beta_2$$

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$$\therefore PM = AM \cot \frac{\beta}{2}$$

$$= \frac{i}{2} (z_2 - z_1) \cot \frac{\beta}{2} \quad (2)$$